

Embedded Operating Systems

Subject Code: R23CI2606

Semester 4 · 5 Credits · Theory · CWIT R23

2 Curated YouTube Lectures

Comprehensive Study Notes with Curated Lectures, Study Guide & Practice Questions

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Coverage Focus

Real-time systems, tasks, scheduling, queues, semaphores, mutexes, interrupts and FreeRTOS basics.

Curated YouTube Lecture Resources

These playlists have been curated to match your CWIT R23 syllabus. Start with the Primary playlist, then use Alternate playlists for different explanations.

Primary Lecture 1: RTOS Fundamentals and FreeRTOS

Channel: *Embedded systems playlist* · Language: *en*

URL: https://www.youtube.com/playlist?list=PLmdxyCs9ZImdo9VA_Cg7rbJpWLtNCYcZ2

Introduces real-time concepts and practical FreeRTOS configuration and task management.

Alternate Lecture 1: FreeRTOS for Microcontrollers

Channel: *Learn Embedded Systems* · Language: *en*

URL: <https://www.youtube.com/playlist?list=PL8IID45FIDTQz6Du4nUfaqJvxOwz86xik>

Hands-on task, queue, semaphore and mutex examples for microcontroller boards.

Study Guide

Embedded Operating Systems (R23CI2606) is a theory subject worth 5 credits. Coverage focus: Real-time systems, tasks, scheduling, queues, semaphores, mutexes, interrupts and FreeRTOS basics.

Key Topics to Study

- Real-time systems
- tasks
- scheduling
- queues
- semaphores
- mutexes

How to Study This Subject

- Watch the primary YouTube lecture playlist first
- Take notes while watching — write key terms and formulas
- If a topic is unclear, watch the alternate playlist
- For programming subjects, type out every code example yourself
- Use Ask LEO to clarify any doubts
- Practise with quiz questions after each topic

Exam Tips

- Focus on the coverage focus topics — these align with your CWIT R23 syllabus
- Review the primary lecture playlist before exams
- Practise numerical problems and code examples
- Time yourself while practising

Subject Information

Field	Value
Subject Code	R23CI2606
Subject Name	Embedded Operating Systems

Field	Value
Semester	4
Credits	5
Category	Theory
Scheme	CWIT R23
YouTube Resources	2

All YouTube Resources

#	Title	Role	Channel
1	RTOS Fundamentals and FreeRTOS	Primary	Embedded systems pla
2	FreeRTOS for Microcontrollers	Alternate	Learn Embedded Syste

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These notes are supplementary to your classroom learning. Always follow your teacher's instructions and the official CWIT syllabus.